

Utkarsh Singh

Pune, Maharashtra | +91-7355207641 | utkarsh20052002@gmail.com | linkedin.com/in/utkarshsingh2002 | github.com/UTKI20

Professional Summary

Computer Science undergraduate (9.36 CGPA) specializing in Applied AI, currently an **AI/ML Intern at Tata Motors**. My work spans **Generative AI**, **NLP**, **RAG pipelines**, and **Deep Learning** across healthcare and scientific discovery. Seeking AI/ML Engineer and Data Science opportunities.

Education

Bachelor of Technology, Computer Science and Business Systems	July 2023 – June 2027 (present)
Bharati Vidyapeeth Deemed University, Pune	CGPA: 9.36
Higher Secondary (Class XII): PCM & Psychology	April 2015 – March 2023
City Montessori School, Lucknow	XII: 94% X: 91.8%

Experience

Tata Motors | AI/ML Intern Present

- Designing an AI-assisted **computer vision** framework for automated engine oil seal inspection, integrating **camera image acquisition** with quality inspection measurements.
- Building a centralized **inspection data pipeline** and **database schema** for **AI-ready training records**, and architecting the end-to-end **machine learning workflow** from preprocessing to deployment.

Skills

Languages: Python, SQL, JavaScript, TypeScript
Libraries: TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy, OpenCV, Hugging Face Transformers, SHAP, ChromaDB, SMOTE
Web & Databases: React, Next.js, Node.js, FastAPI, PostgreSQL, Supabase
Tools: Git, GitHub, VS Code, Streamlit Cloud
Certifications: OpenAI GPT-4, Intro to Data Science, Information Security, NLP: Infosys Springboard;
Python Programming: Coursera

Projects

Sinter(AI Perspective & Counter-Argument Engine) | Github March 2026–May 2026
Node.js, React, PostgreSQL (Supabase), Llama-3, RAG Architecture

- Architected a **RAG** Chrome extension with **Llama-3.3-70B** as inference engine, fetching **5** sources/query, deduplicating into **8** snippets, generating grounded counter-arguments from **5,000-character** context.
- Replaced hallucinated citations with live **Serper API** retrieval (**100%** verifiable URLs); designed **PostgreSQL** arbitration system with **caching**, cutting latency **6s** to **<100ms**.

exHUMA:AI Exoplanet Detector | Deployment Link January 2026–February 2026
TensorFlow, CNNs, SHAP, SMOTE

- Engineered a Multi-Branch **CNN-FFT** pipeline ranking **5,087 stars** into **Top-20 candidates**, retrieving 4/5 confirmed exoplanets (**Recall@20: 80%**) via automated **BLS Phase Folding** and **SNR vetting**.
- Applied **SMOTE** with custom class weighting to isolate faint transits against a **<1%** positive rate; integrated **SHAP** to generate **XAI heatmaps**, improving survey efficiency by **80%**.

Manas Mitra:Multilingual Student Support Chatbot | Collaborative | Deployment Link December 2025
Next.js, FastAPI, Gemini API, RAG (ChromaDB), SentenceTransformers

- Architected a clinical **RAG pipeline** using **ChromaDB** to map user queries to **25 distinct CBT frameworks** via **Gemini 2.0**.
- Engineered a zero-latency **semantic routing layer** in **Next.js**, acting as a circuit breaker to instantly route acute distress queries to national helplines while bypassing LLM API dependencies.

Leadership & Achievements

- Organized a **200+** participant prompt engineering competition at Bharatiyam 2026 as **CSBS Deputy Joint Secretary**.
- Secured **Top 30** out of **12000+** teams in **Innervive 9** National Hackathon in the track of AI/ML.